C20-EE-502

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BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DEEE – FIFTH SEMESTER EXAMINATION

ELECTRICAL MACHINES—III (AC MOTORS AND DRIVES)

Time : 3 Hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List out any three advantages of skewing of rotor slots.

2. Define slip in induction motor.

3. What is the need of auxiliary winding in a single-phase induction motor?

4. List out the applications of shaded pole induction motor.

5. Draw the vector diagram of synchronous motor working on under excitation and indicate the vectors and angles in it.

6. What are the main features of synchronous motor?

7. What are the applications of synchronous motor?

8. List out any three advantages of electric drive.

9. What is the use of fly wheel?

10. What is plugging method of braking for shunt motor?

PART—B

8×5=40

Instructions : (1) Answer **all** questions.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* Explain the constructional details of three-phase induction motors.

(OR)

(b) Explain briefly the working of DOL starter with a neat sketch.

12. *(a)* A 12 pole, 50 Hz, 3-phase Induction motor has rotor resistance of $0 \cdot 2\ \Omega$ and standstill reactance of $0 \cdot 25\ \Omega$ per phase and it is running at a speed of 485 rpm on full load. The rotor induced emf per phase at standstill is 35 volts. Calculate *(i)* starting torque, *(ii)* full load torque, *(iii)* maximum torque and *(iv)* speed at maximum torque. Assume any missing data.

(OR)

(b) The power input to a 500 V, 50 Hz, 6 pole, $3\,\phi$ induction motor running at 970 rpm is 40 KW. The stator losses are 1 KW and the mechanical losses are 2 KW. Calculate *(i)* the slip, *(ii)* the rotor copper losses, *(iii)* shaft output and *(iv)* the efficiency. Assume any missing data.

13. *(a)* Explain the construction and working of a single-phase split-phase induction motor with a neat diagram.

(OR)

(b) Explain the construction and working of single-phase universal motor with a neat diagram.

14. *(a)* A motor has the following duty cycle :

(i) Load rising uniformly from 150 KW to 450 KW in 5 seconds

(ii) Constant load 350 KW for 2 seconds

(iii) Regenerative braking power returned to supply from 100 KW to 0 KW in 1 second. The interval for decking before the next load cycle start is 2 seconds.

Draw the load cycle and suggest suitable continuous rated motor.

Assume any missing data.

(OR)

(b) Explain about the need of load equalization and also explain how it can be achieved by using fly wheel.

15. *(a)* A 400 V, 25 HP, 500 rpm DC shunt motor is to be braked by plugging when running on full load. The efficiency of the motor is 74·6% and the armature resistance is 0·2 Ω. Determine the following :

(i) The value of braking resistance necessary if the maximum braking current is not to exceed two times the full load current

(ii) The initial braking torque

(iii) The maximum braking torque

(iv) The braking torque when the motor is just reaching zero speed.

Assume any missing data.

(OR)

(b) Explain the method of rheostatic braking as applied to DC shunt motor.

PART—C

10×1=10

Instructions : (1) Answer the following question.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. Discuss in general how to select a suitable motor for any type of application based on different aspects.

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